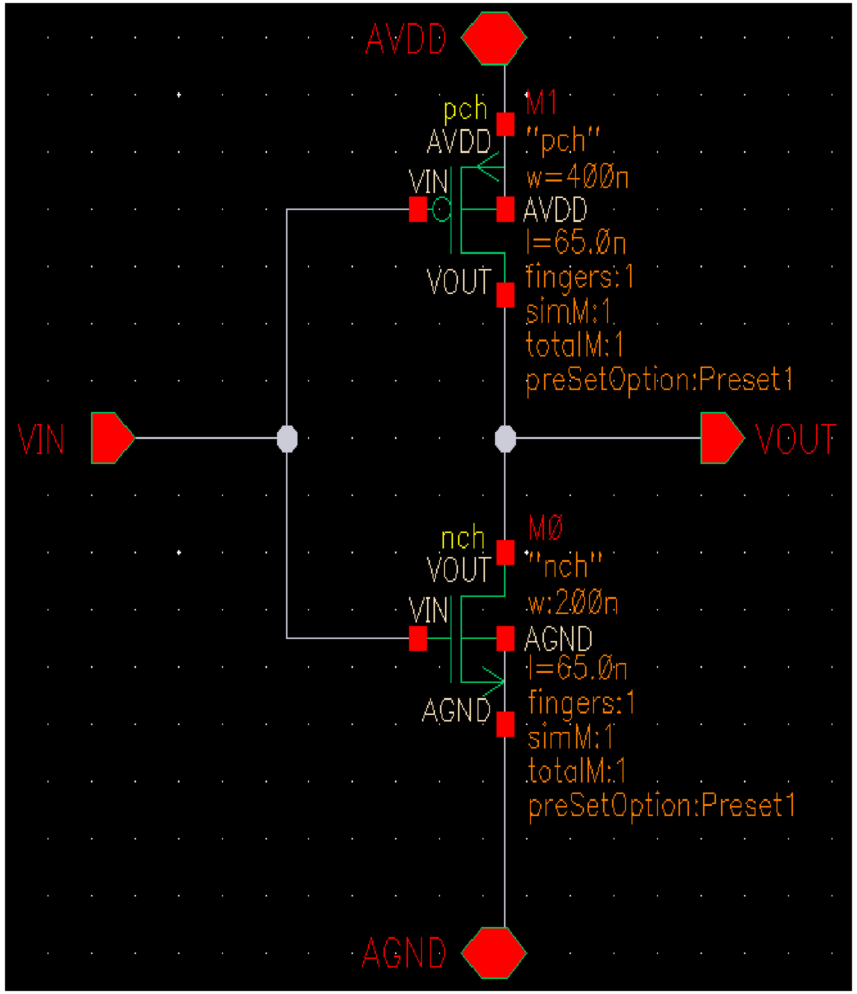
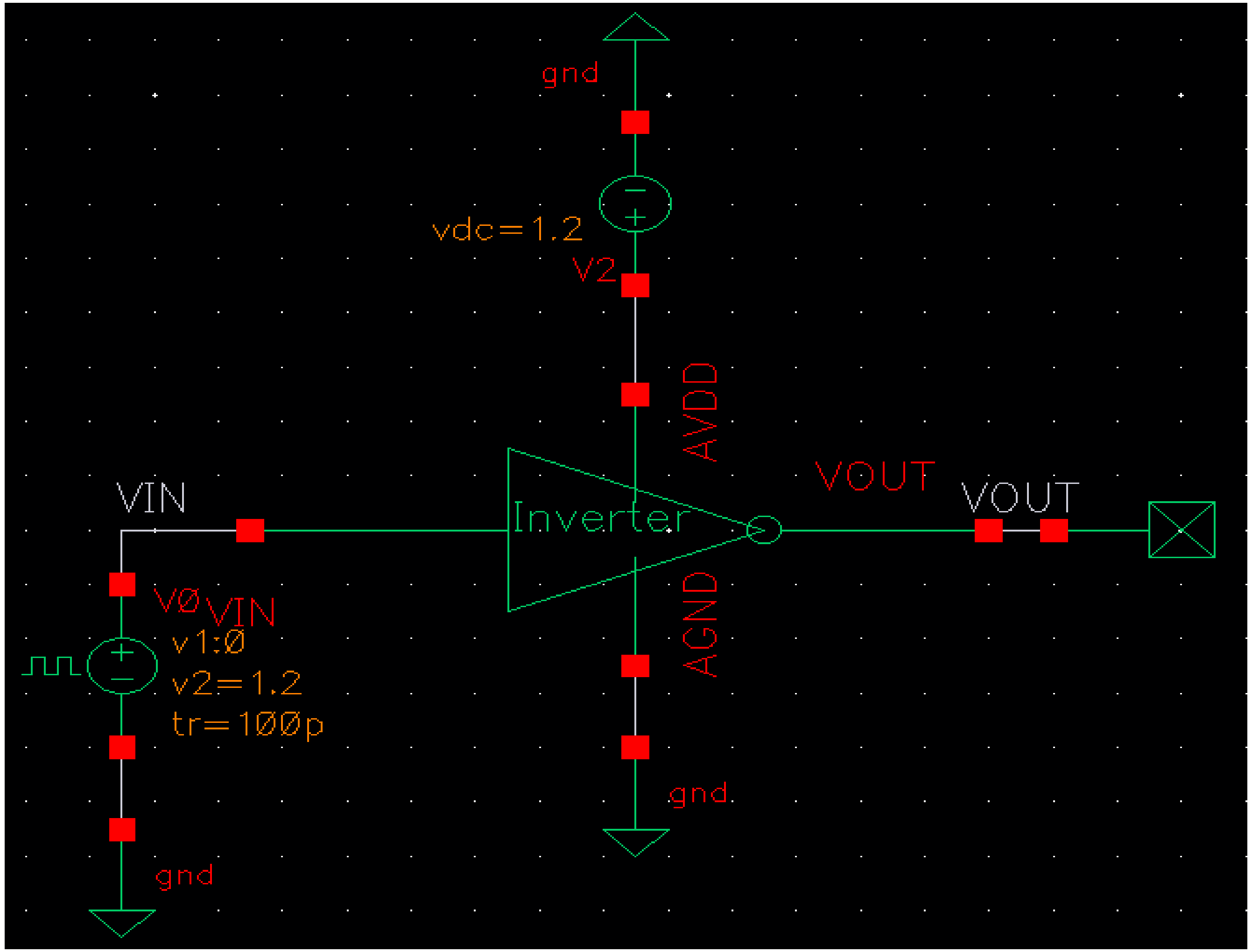
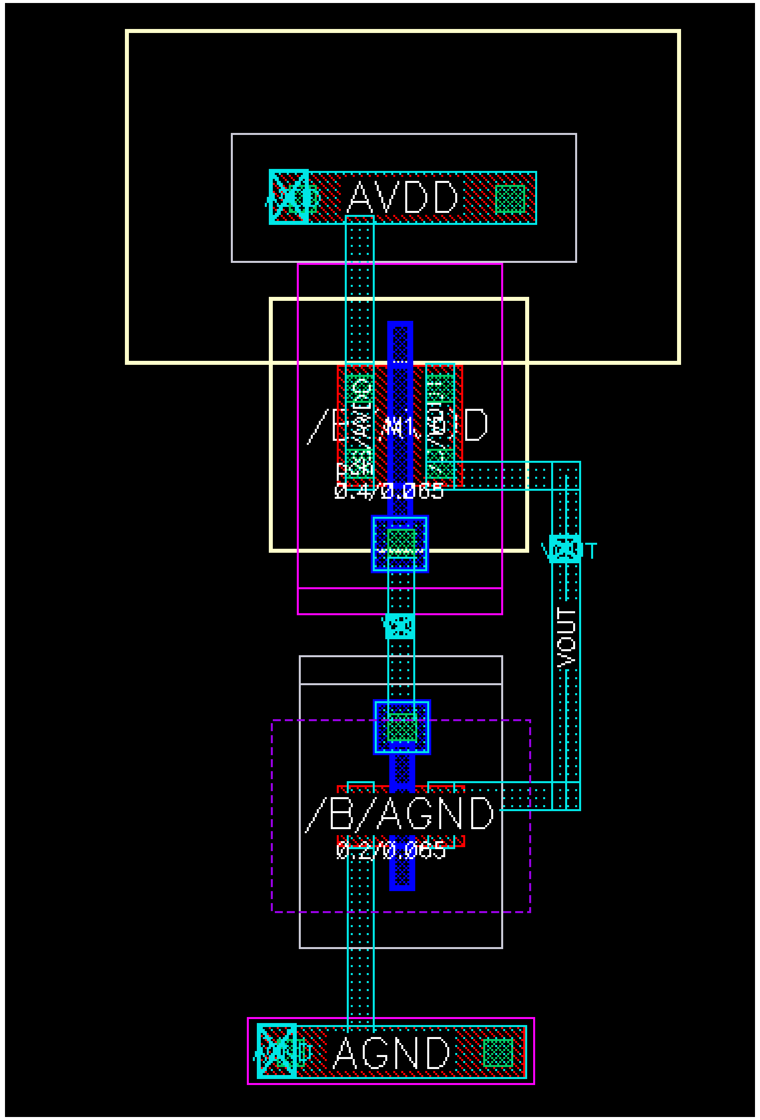
Inverter Layout
using
TSMC 65nm Technology

Inverter Schematic and Test Bench





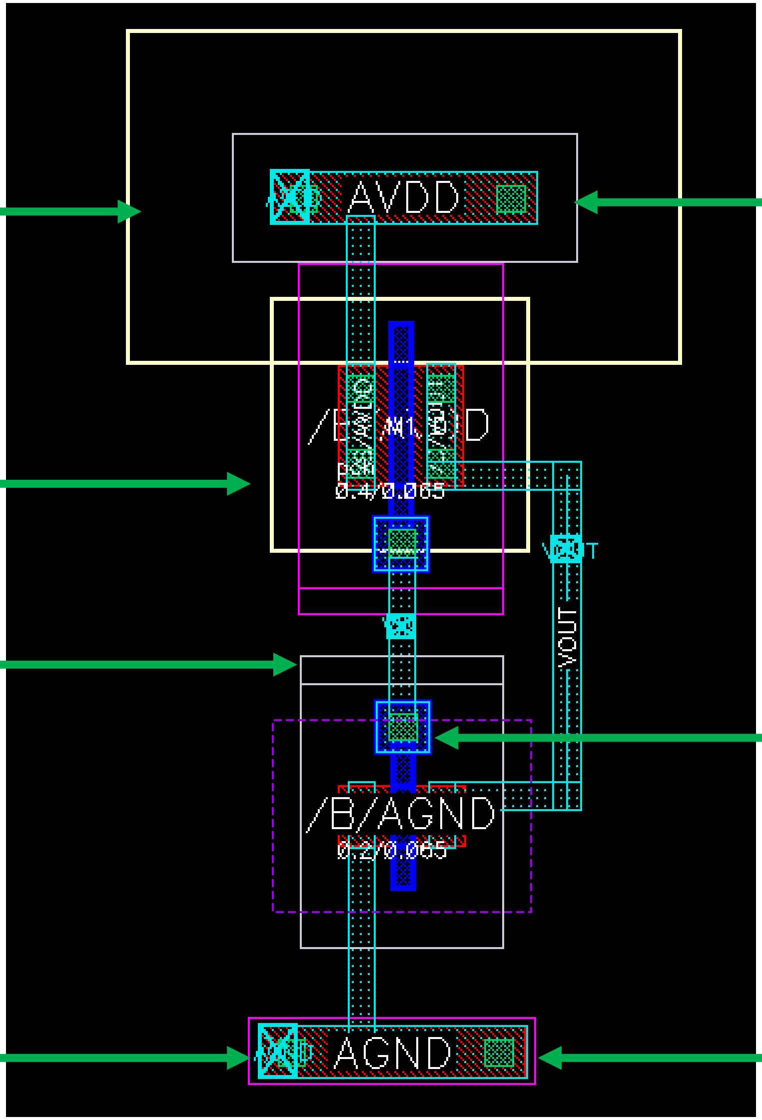
Inverter Layout



Inverter Layout Notes

Nwell (NW) used to avoid LUT effect

Nplus (NP) is added to reduce the resistance contact of the Nwell when it is connected to VDD



Nwell (NW) where PMOS device is fabricated on it . It comes embedded with the PMOS device

Nplus (NP) region is extended to solve poly DRC error . Same extension was made in case of PP

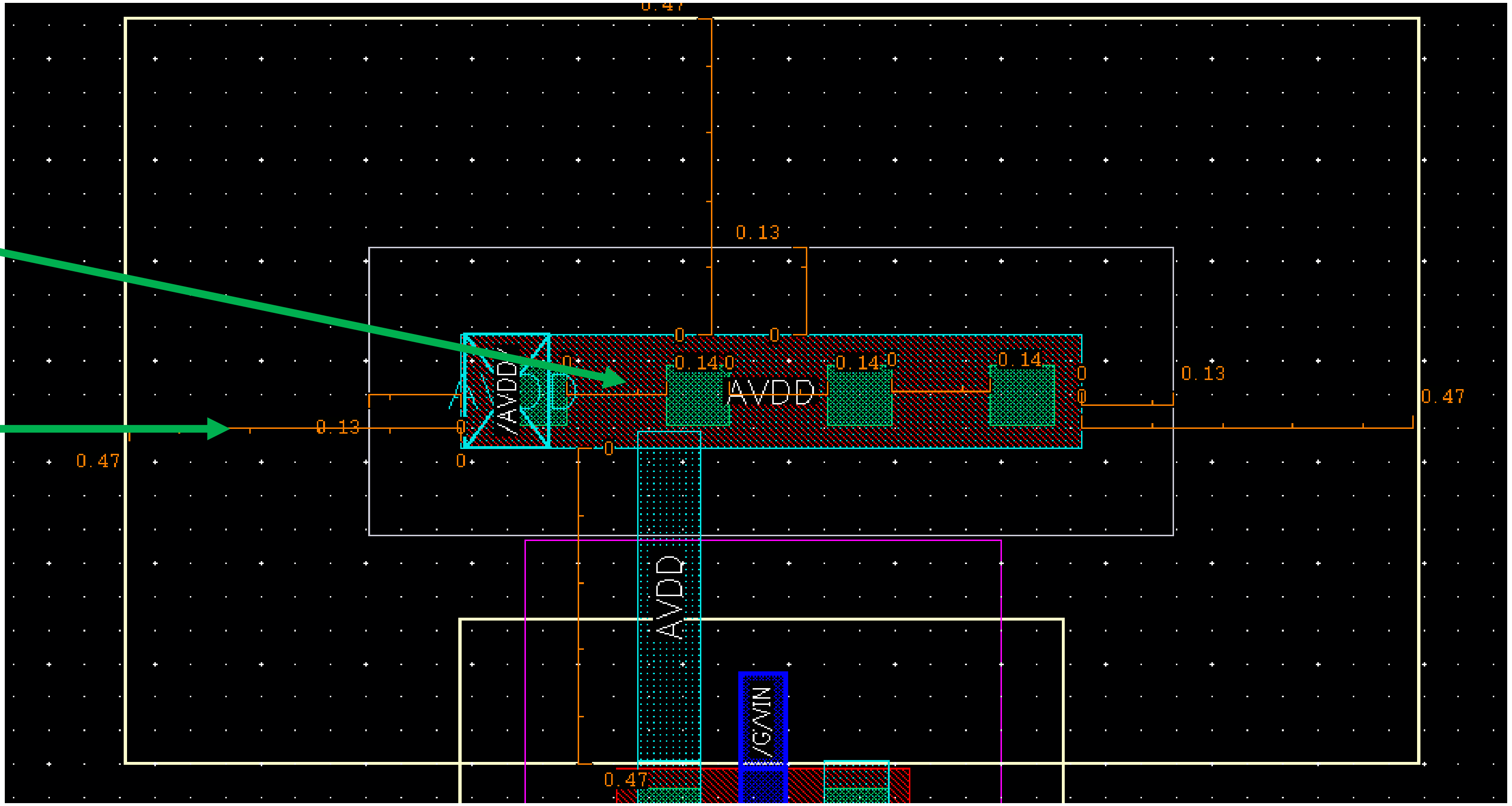
Poly Contact

Oxide Diffusion (OD) layer is made and Contacts (CO) are put on it and metal 1 covers OD

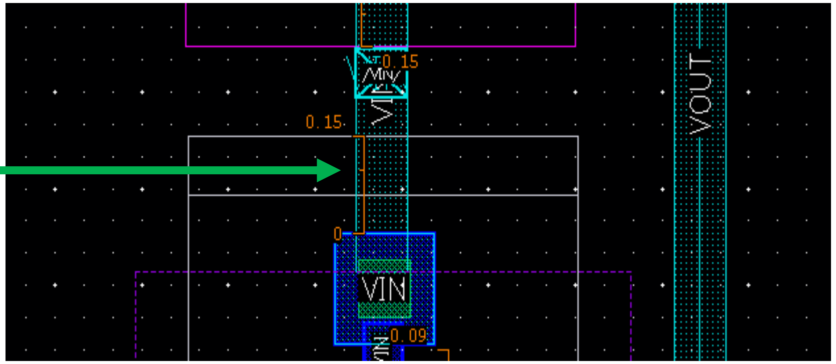
Pplus (PP) is added to reduce the resistance contact of the substrate when it is connected to GND

Inverter Layout Notes (Cont.)

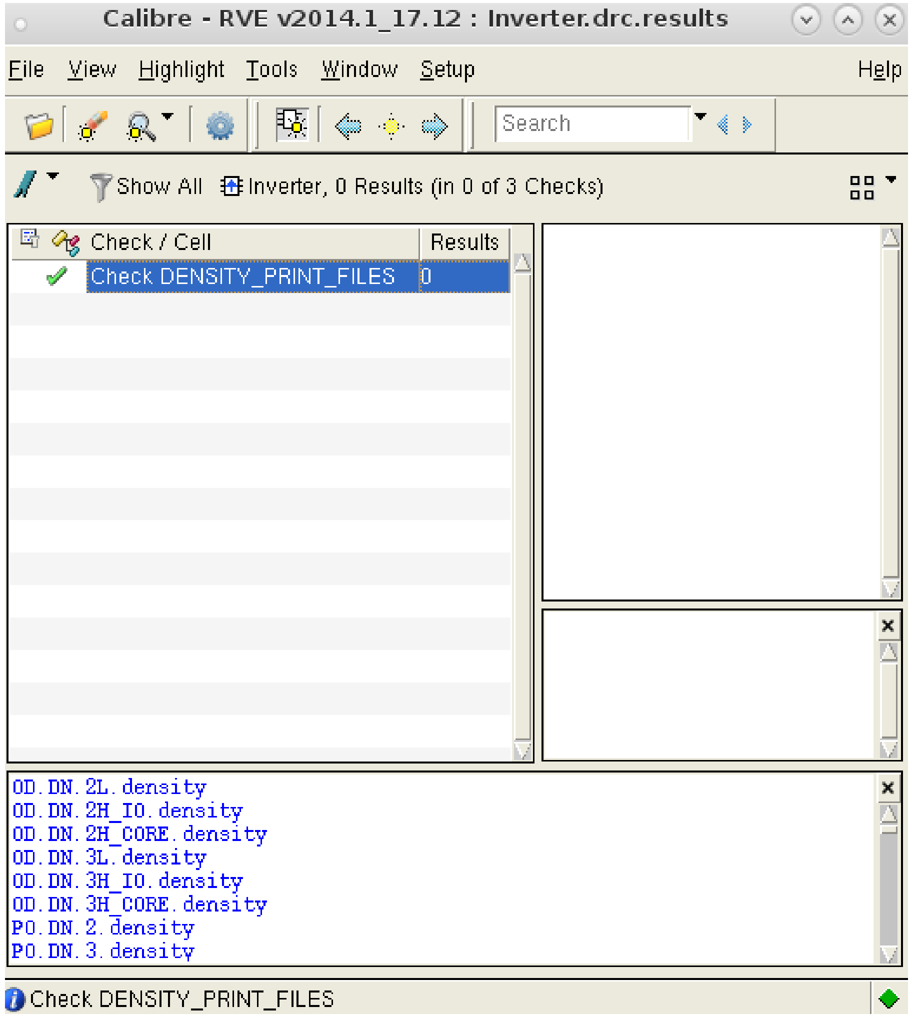
- Spacing beyween contacts and each other 0.14 um
- Spacing between np and OD is 0.13um
- Spacing between OD and Nwell 0.47um

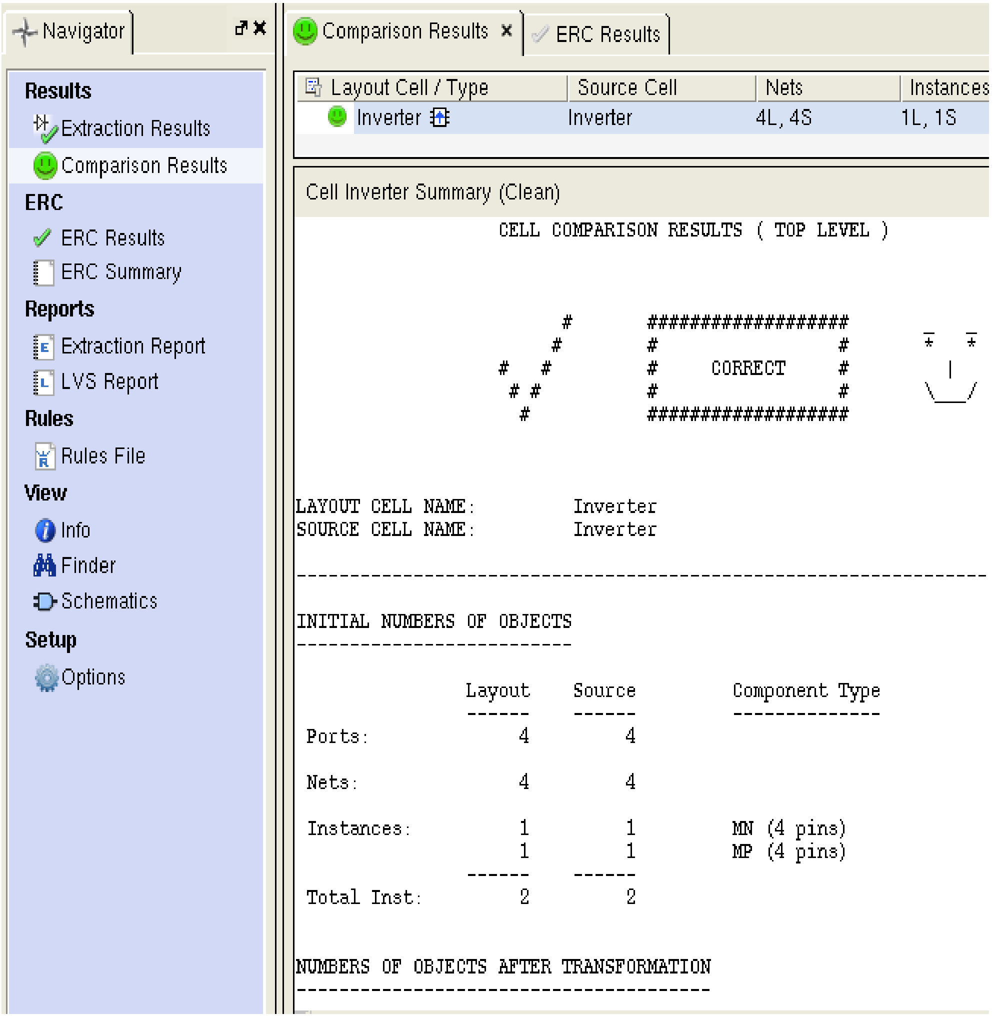


- Spacing between Poly and NP is 0.15 um

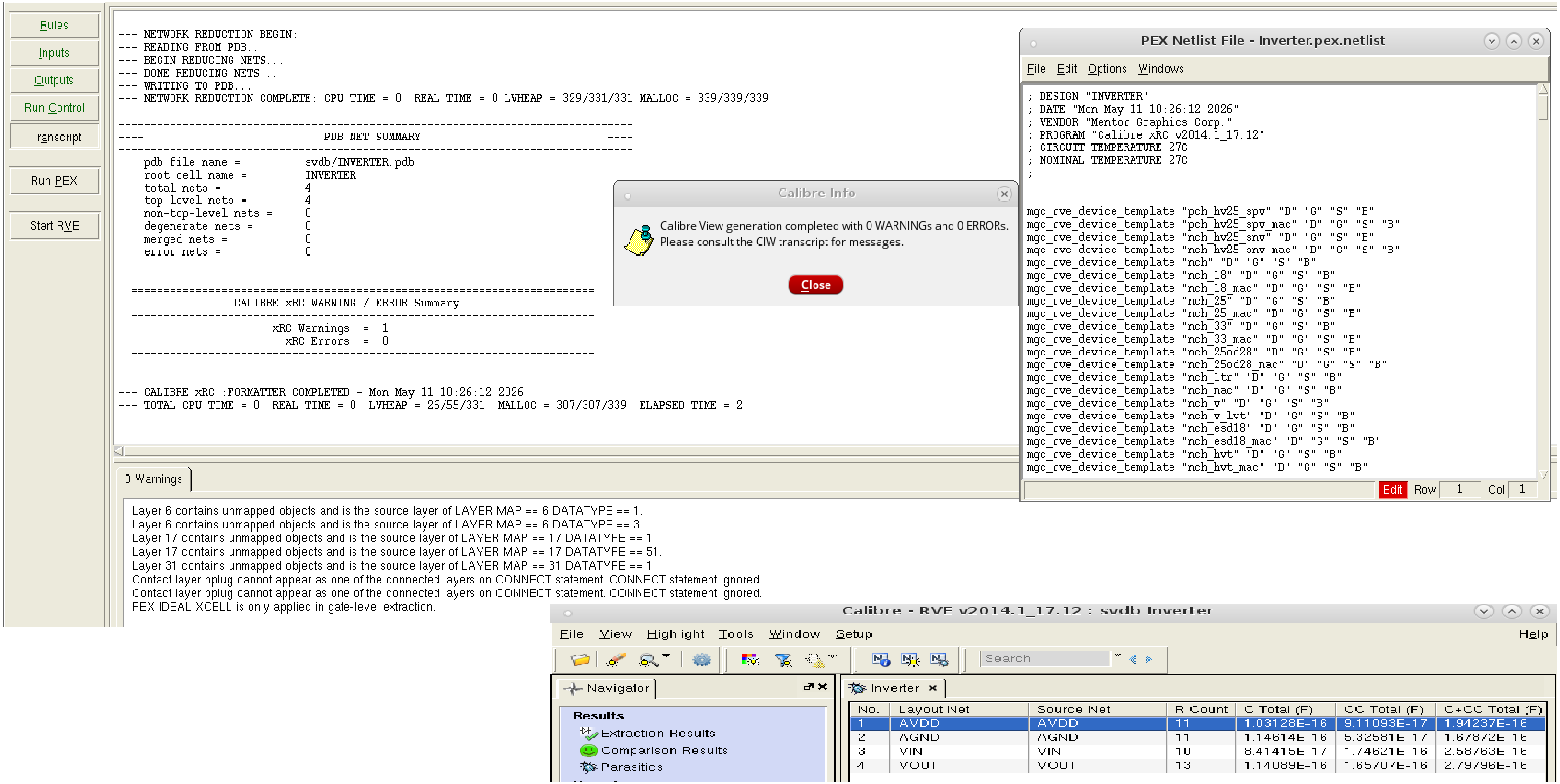


DRC and LVS


Calibre - RVE v2014.1_17.12 : Inverter.drc.results
File View Highlight Tools Window Setup Help
Search
Show All
Inverter, 0 Results (in 0 of 3 Checks)
Check / Cell
Results
Check DENSITY_PRINT_FILES
0
OD.DN.2L.density
OD.DN.2H_IO.density
OD.DN.2H_CORE.density
OD.DN.3L.density
OD.DN.3H_IO.density
OD.DN.3H_CORE.density
PO.DN.2.density
PO.DN.3.density
Check DENSITY_PRINT_FILES


Navigator
Results
Extraction Results
Comparison Results
ERC
ERC Results
ERC Summary
Reports
Extraction Report
LVS Report
Rules
Rules File
View
Info
Finder
Schematics
Setup
Options
Comparison Results
ERC Results
Layout Cell / Type
Source Cell
Nets
Instances
Inverter
Inverter
4L, 4S
1L, 1S
Cell Inverter Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
LAYOUT CELL NAME: Inverter
SOURCE CELL NAME: Inverter
INITIAL NUMBERS OF OBJECTS
Layout
Source
Component Type
Ports: 4 4
Nets: 4 4
Instances: 1 1 MN (4 pins)
1 1 MP (4 pins)
Total Inst: 2 2
NUMBERS OF OBJECTS AFTER TRANSFORMATION

PEX and Result Viewing Environment (RVE)



Simulation Pre-Layout and Post-Layout results